Response to Reviewer FBhK

BioCoLoop, submission 5046 | Revision dated 9 November 2026. Prepared for author review; not posted to OpenReview.

Thank you for identifying two questions that need separate evidence: whether history improves search at fixed access and budget, and whether development-selected changes generalize. We have added the requested random/direct controls, a fresh three-seed history-representation experiment, and a direct diagnosis of the validation failures. The original results remain visible in consolidated tables.

1. Same ten laboratories, configuration library and search budget (Weakness 1; Q1)

Section 4.7 and Appendix E.1-E.2 now compare random search without replacement, GP-EI, and direct/history-guided Qwen and Luna using the same ten training laboratories, ten development panels, twelve configurations, initialization and checkpoint-selection rule. Each search receives six proposal slots beyond D00, with 100 rounds per fit and a 700-round cap. Random search receives the loop's search budget, not the fixed model's single-fit budget.

We verified all 60 task/configuration fits. Enumerating all 462 six-design subsets gives the exact expected outcome of random search conditional on this cache, with selection based only on development evidence. This is not 462 independent training repetitions. Shared-first-proposal and realized-fit-count controls additionally account for failed LLM proposals; for example, four-valid-proposal random search reaches 26.30 on Tahoe versus Qwen's 25.37 under the same realized fit count.

The comparison identifies a positive case without implying a universal gain: Norman/Luna history reaches 23.11 Top-1, versus 22.30 for direct/GP-EI and 22.66 for random search. DTI favors direct Luna/GP-EI (94.60 versus history's 94.27); proteomics ties throughout. On VCC, direct/history and GP-EI retain the fixed reference at 31.50, while random search averages 27.53. We report all five endpoints and all controls.

The main access-scenario table answers a different question: what a laboratory gains by participating in collaborative research instead of developing a model with only its own measurements. Broader access adds training data, development evidence and local computation. We now distinguish this workflow comparison explicitly from the same-access search table; it is not evidence for history alone.

2. Fresh history-representation experiment

Appendix E.6 adds 48 newly trained design/seed fits and new paired Qwen2.5-7B-Instruct search runs. Four study-cell contexts from Replogle and Nadig provide 1,208 training genes, 414 development genes and 406 held-out genes, with gene-disjoint partitions. This is a separate mechanism benchmark, not a replacement score for the original cell endpoints.

Direct, raw-history and structured-history search share the first proposal and six-request budget. Raw and structured history observe the same queried development facts; structured cards explicitly organize configuration changes, reference comparisons and expected-versus-observed outcomes. Direct, structured history, random search and GP-EI each use seven full fits, 16,940 optimizer steps and 532,840 training-example exposures per seed. We report additional LLM costs separately. Raw history has three invalid slots across the three runs, which remain charged rather than silently replaced.

Mean held-out Top-1 is 22.625 for direct, 22.625 for raw history, 22.888 for structured history, 22.668 for random search and 23.152 for GP-EI. Structured versus direct yields two gains and one tie, averaging +0.263 points; its paired gene-cluster 95% interval is [-1.022, 1.568]. Thus the revised representation changes candidate selection and has a positive point estimate, but does not establish stable superiority over simpler search. These complete controls define the scope of our history claim.

The separate 10.84-point Norman result compares training allocation with uniform fitting, not LLM history. New promotion controls show that random promotion reaches 22.63 versus 22.32 for the trajectory policy. We retain that comparison and do not use it as evidence of a unique history mechanism.

3. What explains the development-test divergence? (Weakness 2; Q2)

We reconstructed all 14 distinct checkpoints selected in the extended VCC/Tahoe searches, re-evaluated their 140 development panels, and independently reproduced all 80 archived prefix test scores. There is no aggregation mismatch in these cases: each VCC panel contains two development interventions and each Tahoe panel six, so equal-panel macro Top-1 is exactly the pooled intervention-macro score, even when query counts differ.

The development population is small: 20 VCC interventions and 60 Tahoe compounds. Qwen/VCC's +1.17 development points accompanies -11.89 test points; Tahoe's +5.00 development points represents three additional correct development compounds, while test performance decreases by 11.94 points. To test dependence on the option sets, we evaluated the same initial/final checkpoints on 20 paired development-roster redraws, without training or reselection. Their mean advantages become -1.34 and -1.33 points, positive in only 6/20 and 7/20 redraws. This is direct evidence of roster sensitivity, not proof that one particular cause explains all failures. Finite-sample selection instability is a known model-selection concern Cawley and Talbot, 2010.

The larger follow-up retains every development/test prefix for all methods and three seeds. All final search outcomes exceed their fully fitted D00 references, so the earlier large terminal decreases do not recur. A small interim reversal remains: in seed 62, raw and structured history select D05 at slot 2, improving development by 0.057 points but reducing test by 0.257; later selections recover. A context-specific reversal also remains on K562. Because the data, response panel and candidate library differ, we do not attribute this change to feedback formatting alone or claim that selection error is eliminated. The follow-up supplies a new gene-disjoint study, not repeated regroupings of the original VCC/Tahoe partitions.

4. What changed in the manuscript

Appendix E.7 also adds 40 fresh fixed-design fits, rotating the single participating laboratory through all ten identities on proteomics and the three cell endpoints. Laboratory 0 is below the independent-laboratory mean for protein/VCC but is the best on Norman, showing why its identity cannot stand for all laboratories. This one-seed CPU sensitivity is reported separately from the three-seed GPU workflow table; it does not substitute for rotating every controller or for the same-ten-laboratory search comparison.

The original B.10/B.12 outcomes are consolidated into two clearly separated protocol panels; B.11/B.13 trajectory statistics are consolidated likewise. All positive, tied and negative outcomes remain. The targets are explicitly retrospective, and all six or 24 slots were executed: earlier target attainment is not described as realized compute saving. A redundant two-source pilot table is condensed into text while retaining every source-specific result and the absence of a loop gain. Sections 4.7-4.8 and Appendix E now connect the matched controls, fresh experiment and validation diagnosis directly to the two questions raised in your review.

Response to Reviewer AfJB

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Thank you for identifying the relevant prior work and the need to separate model-development benefits from additional laboratory access. We have revised both the positioning and the experimental comparison rather than treating the original main table as evidence for the history mechanism.

1. Relation to existing collaborative research systems

The revised related-work section discusses all three systems explicitly. The Virtual Biotech coordinates specialist agents around biomedical evidence for therapeutic decisions. AutoScientists maintains shared experimental state, including unsuccessful directions, and coordinates subsequent experimentation. Helmsman is directly relevant to automated federated design: its planning, coding and simulation-based diagnosis already connect LLM agents to federated system construction and refinement. We have updated its citation to the ICLR 2026 publication.

BioCoLoop's contribution is the biological model-development interface and its controlled evaluation. Task adapters connect interaction, observed-response proteomic and perturbation predictors to the same local-fitting, development-evaluation and proposal interfaces. The coordinator distinguishes training-example-weighted parameter updates from equal-panel development evidence. Source-specific heads allow studies with different response objectives to contribute to a shared representation while the target development population remains explicit. The evaluation then separates laboratory participation, proposal-history conditioning and training allocation.

This is a narrower and more concrete contribution than inventing shared research memory or automated federated design. We now state the four-factor finite search space directly, and do not portray it as open-ended architecture or source-code discovery. The abstract, introduction and conclusion have been rewritten around the implemented framework and the evidence for each decision.

2. Does the history mechanism itself improve search?

We added random search without replacement and fixed GP-EI Bayesian optimization under the same ten-laboratory access as direct and history-guided LLM search. All methods use the same twelve configurations, development panels, initialization and six-proposal cap. The new main-text table reports all five endpoints; Appendix E provides exact random-search variability, shared-first-proposal and realized-fit-count controls.

The result is not a general history advantage. Qwen direct and history select equally scoring models on all five endpoints. Luna history improves Norman from 22.30 to 23.11 Top-1, versus a random expectation of 22.66, but decreases DTI from 94.60 to 94.27 AUROC. GP-EI matches direct Luna on DTI. Proteomics retains the starting configuration, and the VCC score attained by the LLM searches is already attained by the fixed configuration. These controls directly address the alternative explanation that conventional configuration search is sufficient in much of this small design space.

The comparison reuses verified complete candidate fits, with development-only selections sealed before additional scoring. It is a retrospective one-training-seed-per-endpoint study, not new multi-seed evidence. Three-seed source-defined laboratory controls are also reported separately: Qwen history is below direct on both proteomics and cell prediction, and the additional Luna feedback experiment yields only a small protein change and cell ties. We retain these results rather than extending the combined-system gain to unsupported component claims.

3. What the revised empirical contribution supports

The original main table is now explicitly an access-scenario comparison, with a Labs column. The new same-access table addresses search quality. Re-evaluation and new development-roster diagnostics identify validation sensitivities associated with poor held-out generalization.

We also completed 360 fresh fits to compare the allocation policy with uniform, ordinary score-only, learning-rate-stratified score-only and random promotion at identical training/evaluation counts. The original Norman gain over uniform is reproduced, but score-only promotion recovers most of it (21.70 versus 22.32), and the exact random-promotion expectation is 22.63. The trajectory policy improves Tahoe relative to both alternatives, while random is better on VCC. We report these mixed controls in Section 4.4 and Appendix E.5 rather than claiming a generally superior new allocation algorithm. The revised contribution is the implemented biological interface and its controlled empirical assessment, with data-access, search and allocation effects distinguished.

Response to Reviewer cR5g

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Thank you for the concrete concerns about conventional tuning, laboratory heterogeneity and the interpretation of the baseline comparison. We have added the missing controls and reorganized the evidence around those questions.

1. Conventional hyperparameter optimization

The revision adds a same-access table comparing random search without replacement, fixed GP-EI Bayesian optimization, direct LLM search and history-guided LLM search. All use ten laboratories, identical development panels, the same twelve-design library and a six-proposal cap after the initial configuration. Exact enumeration of 462 candidate subsets reports random search's expected outcome and search-policy variability; it does not count these subsets as independent training seeds.

For example, DTI AUROC is 94.60 for GP-EI, 94.45 in expectation for random search and 94.27 for history-guided search. On Norman, Luna history reaches 23.11 Top-1, versus 22.30 for GP-EI/direct and 22.66 for random search. Other endpoints include ties and cases where the fixed configuration is already competitive. Thus the new evidence supports task-dependent search behavior, not general superiority to conventional tuning. We also report controls matching the actual number of valid fits when an LLM proposal fails.

2. Laboratories defined by different studies

The main ten-client experiment partitions one source dataset; it is not ten independently operated institutions. We now make this explicit and separate it from the source-defined laboratory experiments.

The proteomic experiment treats PTPC, Lin, Ruprecht and decryptE as four laboratories. Their different response objectives use source-specific heads and a shared representation; the same target development panels select checkpoints. In the fixed-design transfer comparison, adding the three external studies increases target AP from 34.41 to 37.53.

The cell control compares four same-source VCC groups with VCC, Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562 as four source-defined laboratories. Each arm has four laboratories and 60 training interventions, with the same target anchor. The external-source arm scores 28.82 Top-1 versus 30.70 for the same-source arm. Underlying cell counts and gene panels remain different and are documented. DTI likewise reports each auxiliary source: Davis/Human improve the fixed target model, while BioSNAP and the combined-source setting do not.

These experiments test real differences between released studies, although execution is simulated on one host. We do not present heterogeneous data as automatically beneficial. Appendix E also reports the completed three-seed direct/history controls within the source-defined protein and cell settings, including their negative results.

3. Why direct search can be worse than the fixed model

“Direct” denotes feedback-free LLM configuration search, not training one model once. Every method ultimately deploys one selected predictor; ten laboratories are data holders contributing to fitting, not a ten-model ensemble. Both direct and history search retain the model with the best development score, whereas “fixed” is a single predefined configuration. Retaining a better development model therefore need not retain a better test model.

We investigated this failure rather than changing the reported test selection. Reconstructed VCC/Tahoe predictions show that equal-laboratory and pooled-intervention macro aggregation agree exactly. Small gains on 20 VCC or 60 Tahoe development interventions can accompany much larger test decreases. Re-evaluation under 20 new paired development-option rosters further shows that Qwen/VCC and Tahoe's original development advantages are not stable to distractor/option-order changes. No models were retrained or reselected in this diagnostic.

4. Scope and presentation of the contribution

The original access comparison now has an explicit Labs column and is no longer used to attribute the benefit of additional data to the LLM loop. The same-access search table addresses that attribution directly. We describe the method as a common biological model-development interface with four editable configuration factors, distinct from open-ended model invention. Related work now discusses Virtual Biotech, AutoScientists and Helmsman. A separate 360-fit allocation comparison adds ordinary score-only and random promotion: it reproduces the original benefit over uniform allocation but shows that much of it is also attained by simpler policies. The revised conclusion states these results separately from collaborative fitting and proposal history.